# Balanced Read-Write System Capacity Example

Read-Write Heavy System Example (Trading App)

Assumptions:
- DAU = 1M active traders
- Reads (price lookups) = 200 per user per day
- Writes (orders) = 20 per user per day
- Request size = 300 bytes (read), 500 bytes (write)

Step 1. RPS:
- Reads: (1M × 200) / 86,400 ≈ 2.3K RPS
- Writes: (1M × 20) / 86,400 ≈ 231 RPS

Step 2. Throughput:
- Reads: 2.3K × 300B ≈ 0.66 MB/sec
- Writes: 231 × 500B ≈ 0.12 MB/sec

Step 3. Daily Storage:
- Orders: 231 RPS × 500B × 86,400 ≈ 10 GB/day

Step 4. Peak Load:
- Reads ≈ 7K RPS, Writes ≈ 700 RPS

Key Observation:
- Mix of reads and writes, both must be scaled carefully.